Question 1

```
Begin
    row ← input
    col ← input.
    Declare Seats row[row][column]
    for i=0 to row-1
        for j=0 to col-1
            Seats[i][j] ← input
        endfor
    endfor.
    print seating arrangements.
    for i=0 to row-1
            print row number.
            for j=0 to col-1
                if seats[i][j]=0
                    print _
                else
                    print X
                endif
            End For.
        End For.
    Set totalAvailable =0
    for i=0 To row-1
        set available =0
        for j=0 to col-1
```

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```
        if seats[i][j]=0
            available = available + 1
        Endif.
    Endfor.
    totalAvailable = totalAvailable + available
Endfor.
Print totalAvailable
k ← input
if k<=0 OR k>col
    print invalid input.
else
    m ← row/2
    SbookedRow ← 0
    EbookedRow ← 0
    Sstart seat ← 0
    Estartseat ← 0
    Sdone ← 0
    Edone ← 0
for i=0 to row-1 while (Sdone = 0 OR Edone = 0).
    for i=0 to row-1 while (Sdone =0 OR Edone =0)
        m1 ← m-1
        m2 ← m+1
        if m1>=0
            count=0
```

```
for j=0 to clo-1
if seats[m1][j]=0
    count<=count+1
else
    count<=0.
Endif.
if count>=k
    SbookedRow<=m1
    Sstartseat<=j-k+1
    Sdone<=1
    Break.
  Endif
 Endfor.
Endif
if m2 <=row
  count<=0
  for j=0 to col-1
    if seats[m2][j]=0
      count=count+1
    Else
      count=0
    Endif
  if count=k
  SbookedRow=m2
```

Edone = 1
Break
Endif
Endfor
Endif
EndFor
Declare bookedrow and StartSeat
If Sdone = 0 AND Edone = 0
print no available seats
Else
if (EbookedRow - m) = (m - SbookedRow)
bookedRow = SbookedRow
StartSeat = SstartSeat
Elseif (EbookedRow - m) > (m - SbookedRow)
bookedRow = SbookedRow
~~StartSeat = Sta~~
startSeat = SstartSeat
Else
bookedRow = EbookedRow
startSeat = EstartSeat
Endif
print booking successfull
print ~~Booked~~ bookedRow

~~for j= startseat to startse~~

for j= startSeat to startSeat+k-1

print j

Seats [booked Row][j] = 1

~~End for~~

Endfor.

print updated arrangend.

print seat numbers.

for i=0 torow-1

print row number

for j=0 to col-1

if seats [i][j]=0

print _

else

print X

Endif.

Endfor.

Endfor

End if

Endif

Stop.

Input

- Number of rows
- Number of columns
- Initial seating arrangement
- Number of consecutive seats required by the customer

Output

- Initial seating arrangement
- Total number of available seats
- Whether the booking is successful or no seats are available
- Row and seat positions allocated to the customer.
- Updated seating arrangement after booking

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